

LESSON 4-3 • SESSION 1 • PRACTICE • FORM A**Estimate with Benchmarks**

Learning target – I can estimate a percent by comparing an amount to benchmark percents. • 6.RP.A.3c

REMEMBER

- A **benchmark** is an easy percent: 10%, 25%, 50%, 75%, 100%.
- First name **the whole** (100%). Then find benchmark amounts: half for 50%, $\div 4$ for 25%, $\div 10$ for 10%.
- Choose the benchmark amount **closest** to the part. For the percent left, subtract from 100%.

WORKED EXAMPLE

A game download is 80 MB. 41 MB are done. About what percent is done?

Step 1 Whole: 80 MB is 100%.

Step 2 50% of 80 is 40 MB.

Step 3 41 MB is just over 40 MB.

Answer about 50%

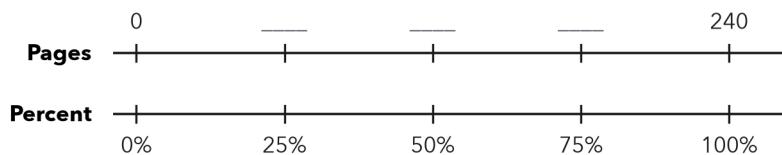
WORDS **estimate** (*estimar*) • **benchmark percent** (*porcentaje de referencia*) • **the whole** (*el todo*)

PRACTICE

show your work • label units

- About what percent? Name the closest benchmark: 10%, 25%, 50%, 75%, or 100%.
 - 41 out of 80 $\approx$ ____ %
 - 31 out of 120 $\approx$ ____ %
 - 19 out of 40 $\approx$ ____ %
 - 47 out of 64 $\approx$ ____ %
 - 13 out of 120 $\approx$ ____ %
 - 58 out of 60 $\approx$ ____ %
- Reading Goal.** Ana's goal is 240 pages this month. She has read 118 pages. Fill in the double number line.

Hint Each jump on the top line is $240 \div 4$ pages.



- Fill in the missing page numbers.
 - About what percent of her goal has she read? _____ %
- Class Fundraiser.** A class wants to raise \$400. So far it has raised \$310.

Hint Find 75% of \$400 and compare.

 - 10% of the goal _____
 - About what percent? _____ %
 - Gym Water Jug.** A water jug holds 32 cups. It has 17 cups left.
 - Half of the jug, in cups _____
 - About what percent is left? _____ %
 - Trail Progress.** A hiking trail is 36 km long. A hiker has walked 27 km.

Hint Find one fourth of 36 km first.

 - About what percent is walked? _____ %
 - What percent is left? _____ %

Show your work.

6. **Estimate Check.** Kwame says 19 out of 60 is about 50%. Is he right? Use 10% of 60 to find a better estimate.

Hint Find 50% of 60 and 10% of 60.

Show your work.

Better estimate _____

7. **Parking Lot.** A parking lot has 80 spaces. 61 are taken. Estimate the percent taken and name the benchmark amount you used.

Show your work.

About _____

8. **Chorus Tickets.** A chorus printed 480 tickets for its concert. It has sold 47 of them.

Hint Divide 480 by 10 to find 10%.

- a. About what percent are sold? _____ %
b. About what percent are not sold? _____ %

Show your work.

FIND THE ERROR

explain the mistake, then fix it

9. **Elena's Work.** A drive holds 200 GB. Elena has used 150 GB, and 30 GB of that is music. She says music is about 75% of the drive. Explain her mistake. Then estimate the percent of the drive that is music.

TEST PRACTICE

choose the best answer

10. 10% of an amount is the same as –
A one half **B** one tenth **C** one fourth **D** ten times it
11. 31 out of 120 is about –
A 10% **B** 25% **C** 50% **D** 75%

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

\$40 E	10% S	75% A	\$4 Z	16 R	90% T	25% E	80% X	50% N	47% Q
#2b		#3a	#3b	#4a	#5b	#8a	#8b		

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____